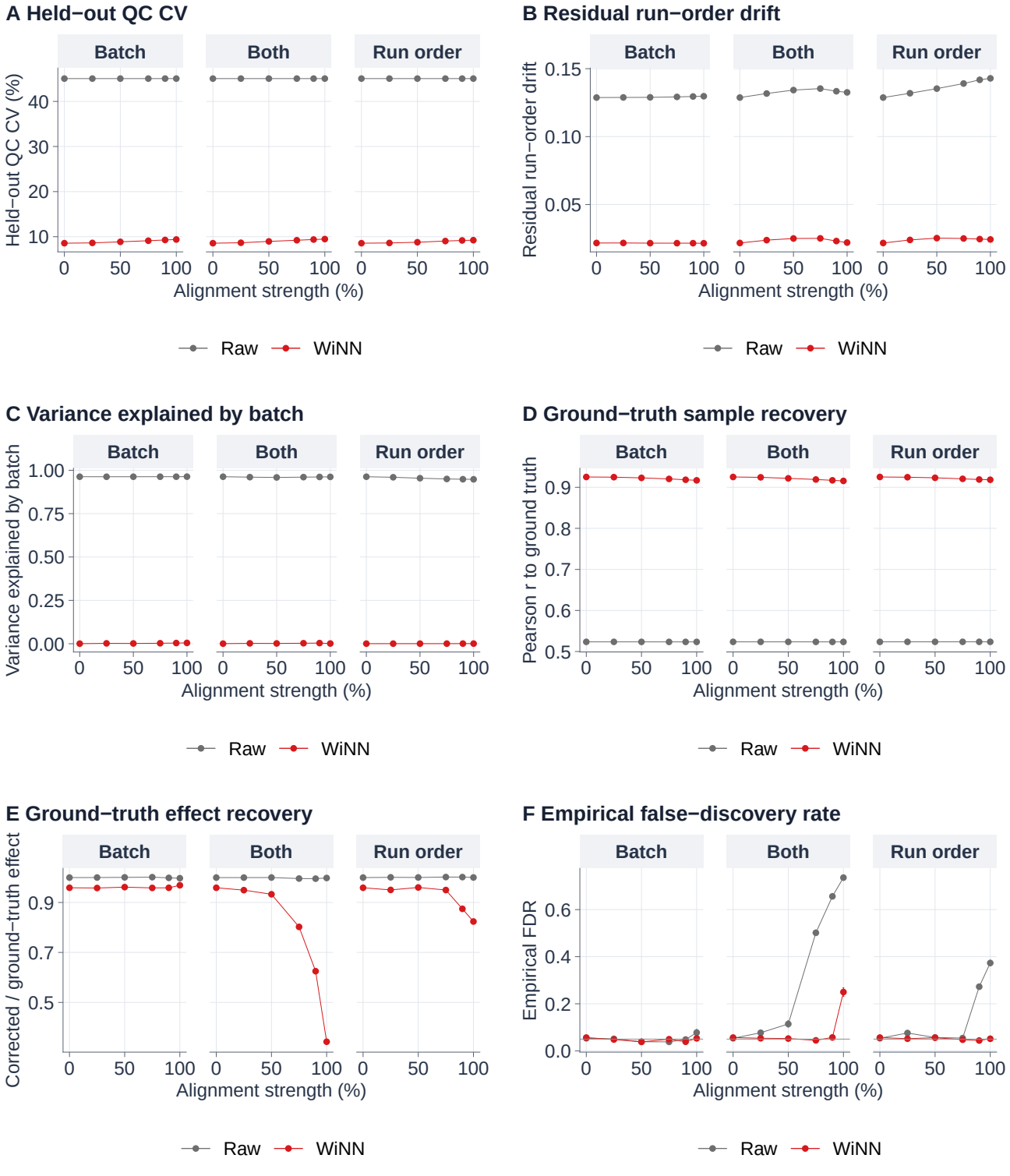

A Held−out QC CV
Batch
Both
Run order
Held−out QC CV (%)
40
30
20
10
0
50
100
Alignment strength (%)
Raw
WiNN
B Residual run−order drift
Residual run−order drift
0.15
0.10
0.05
C Variance explained by batch
Variance explained by batch
1.00
0.75
0.50
0.25
0.00
D Ground−truth sample recovery
Pearson r to ground truth
0.9
0.8
0.7
0.6
0.5
E Ground−truth effect recovery
Corrected / ground−truth effect
F Empirical false−discovery rate
Empirical FDR
0.6
0.4
0.2
0.0